Resume Project Demo

CodePilot AI - Autonomous Software Engineering Agent

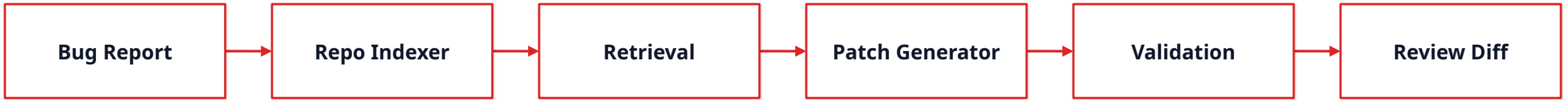
Bug report analysis, code retrieval, patch generation and validation

Built to explain backend engineering, data structures, failure handling, APIs and system design in an SDE I interview.

Demo-ready
Code + Tests + Docker + PDFs

Problem and Architecture

Bug fixing requires reading reports, finding code, preparing patches and validating changes before review.



The design separates request handling, core logic and reliability concerns. That makes it easier to explain scaling and failure handling during interviews.

Key idea

Keep the demo simple enough to run locally, but design it like a real backend service with production concerns.

Core Engineering Features

What this project proves on your resume

- Repository indexing
- Relevant code retrieval
- Unified diff generation
- Automated validation checks
- Human review status

This project is useful because each feature maps to a real SDE interview topic: data structures, APIs, concurrency, testing, state management, and scale.

Use the feature list to lead the explanation. Do not say only "I built an app". Say what engineering problem each feature solves.

API	**Tests**	**Demo**	**Deploy**
REST	pytest	Browser	Docker

Live Demo Flow

A simple sequence to show the project confidently

1. Submit divide-by-zero bug
2. Show retrieved file
3. Show generated diff
4. Show validation passed

Presenter script

1. Start with the problem.
2. Show the architecture.
3. Run the browser demo.
4. Open the API response.
5. End with what you learned.

Command

```
pip install -r requirements.txt && uvicorn app.main:app --reload --port 8000
```

Interview Talking Points

Use these points to answer project questions clearly

1. Engineering workflow matters
2. Retrieval narrows context
3. Validation reduces risky patches
4. Human approval remains the final gate

Resume impact

This project is not just a CRUD app. It shows backend correctness, system design thinking, automated tests and demo ownership.

Best closing line: "I can extend this with Redis, Kafka/SQS, real database migrations, metrics, tracing and AWS deployment."